AQI Forecasting Lab: Project Overview

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AQI Forecasting Lab predicts daily air quality for Mumbai (and Delhi) 1 to 7 days ahead, using classical time series models, and shows every step of the analysis in a local web dashboard.

The problem

Air pollution in Indian cities swings sharply with the seasons, and people only see today's number. A short-range forecast helps residents, schools and researchers plan around bad-air days. The academic question was simple: can a city's own AQI history, analysed with the full Applied Time Series workflow (11 required steps), forecast the next few days better than "tomorrow will look like today"?

What we built from scratch

1. **Picked and verified the data.** Audited 5 candidate sources and chose official CPCB daily AQI bulletins (2015 to 2025, 297 cities). An existing GitHub notebook was reviewed and rejected: it averaged all cities into one series.
2. **Cleaned it properly.** One city at a time on a full daily calendar; modelling starts in 2019, when Mumbai's monitoring network grew from 1 station to 7+.
3. **Ran the full 11-step workflow:** EDA, missing values, ADF and KPSS stationarity tests, log transform, seasonal decomposition, ACF/PACF, lag selection, AIC/BIC.
4. **Compared 9 models fairly:** naive baselines, Holt-Winters, AR, MA, ARMA, ARIMA, SARIMA and SARIMAX with annual Fourier terms. Chosen on 2024, tested on untouched 2025 data, with a significance test against naive.
5. **Deployed it** as a local dashboard (FastAPI + Plotly), plus an executed Jupyter notebook and automated tests.

Requirement checklist

All 11 steps of the course's model deployment workflow are covered, each in its own numbered dashboard section.

#	Requirement	Status	What we did
1	Data collection	Done	Official CPCB daily AQI bulletins, 2015 to 2025, 297 cities; data card with source, licence and coverage
2	Exploratory data analysis	Done	Line graph, scatter plot, histogram, QQ plot and descriptive statistics, plus monthly and weekday profiles and outliers
3	Missing values	Done	173 missing days found; three filling methods compared; gaps of 3 days or less interpolated
4	Stationarity, ADF test	Done	ADF on five versions of the series, rolling mean and SD charts, KPSS as a cross-check
5	Transformations	Done	Log transform (Box-Cox $\lambda = 0.03$), daily and 7-day differencing, tests re-run after each
6	Seasonality and decomposition	Done	MSTL with weekly and yearly cycles, plus classical monthly decomposition
7	ACF and PACF plots	Done	For log AQI, its daily change and the residual, plus a 400-lag ACF
8	AR, MA, ARMA, ARIMA, SARIMA	Done	All five fitted and compared on 2025, alongside naive, Holt-Winters and SARIMAX-Fourier
9	Identify lags	Done	Order search per family guided by the PACF; AR settled on 10 lags
10	AIC and BIC	Done	Table of all 75 fitted models and AIC/BIC-vs-lag charts
11	Deploy the model	Done	Live local dashboard with forecasts and CSV download, one-click launch, public GitHub repo

On step 5: ADF already finds log AQI stationary, so differencing was not strictly required. The log transform was still applied because variance grows with the pollution level, and differenced models (ARIMA, SARIMA) were kept to show the effect empirically.

What the product can do

- Forecast daily AQI 1 to 30 days ahead for Mumbai or Delhi, with 80% and 95% prediction bands and the CPCB category for each day.
- Switch between any of the 9 trained models and download the forecast as CSV.
- Walk a professor or reader through every workflow step, with charts, test results and a plain-language reading of each.
- Show honest accuracy on a year the models never saw: error metrics, interval coverage, category hit rate and significance against naive.
- Run on any laptop with one double-click (`run.bat` on Windows, `run.sh` on macOS/Linux); light and dark themes, works on phones.

Forecasts are study estimates, not official CPCB warnings or health advice.

Key results (2025 test year)

The models clearly beat naive for Delhi; for Mumbai the gain is real but small and not statistically significant.

City	Best model on test	1-day RMSE vs naive (AQI)	7-day RMSE vs naive (AQI)	Better than naive? (Diebold-Mariano)
Mumbai	SARIMAX + Fourier (K=2)	14.8 vs 15.5	26.6 vs 29.7	Not significant (best p = 0.07)
Delhi	SARIMAX + Fourier (K=4)	34.9 vs 38.4	46.2 vs 65.1	Yes, p < 0.0001

Other findings: log transform justified (Box-Cox λ = 0.03); annual seasonality dominates (strength 0.78) while weekly is weak (0.05); 95% intervals cover 97 to 98% of real days; residuals of the selected Mumbai model pass the Ljung-Box test.

Reading the dashboard

The dashboard is one long page. The left sidebar (a row of buttons on phones) jumps to each section, numbered to match the professor's 11 requirements. The city picker at the top right switches every chart and number between Mumbai and Delhi; the moon/sun button switches light and dark themes. All figures below are for Mumbai.

Forecast (Requirement 11: deployment)

The top of the page is the working product.

- **Four headline numbers.** *Last observed*: the most recent CPCB value (130, Moderate, 31 Dec 2025). *Selected model*: the model that won on 2024 data, AR(10). *Test RMSE, 1 day*: its typical next-day error on 2025 (15.2 AQI points) and how that compares with the naive forecast. *95% interval coverage*: how often the real value fell inside the shaded band in 2025 (97%).
- **Controls.** *Model* picks any of the 9 models; *Horizon* sets 1 to 30 days ahead; *Download CSV* saves the forecast with its bands.
- **Forecast chart.** Black line = the last 120 days of real AQI. Blue line with dots = the forecast. Dark blue band = 80% prediction interval, light band = 95%. The faint coloured stripes behind are the CPCB categories (green Good up to red Severe). The dotted vertical line marks the last real observation.
- **Forecast table.** Each future day with its median AQI, its 95% range, and its CPCB category.

Accuracy was only tested up to 7 days. Past that, forecasts drift towards the usual seasonal level and the band widens.

01 Data collection

- **Data card:** source, file, licence, coverage (2015 to 2025, 297 cities, 471,015 rows), columns, units, and the train/validation/test split.
- **Monitoring stations per year:** bars show how many stations fed Mumbai's city average (1 until 2018, then 7, rising to 24). Blue bars are the years used for modelling. This is why modelling starts in 2019: before that, "Mumbai AQI" meant one station.
- **Full history:** every daily value since 2015; the shaded area is the modelling window.

02 Exploratory analysis

- **Summary numbers:** mean 104.2, median 89, standard deviation 54.9, range 25 to 381 over 2,557 days. Skewness 0.83 means a long tail of bad days; on the log scale it drops to 0.01 (almost symmetric).
- **Distribution:** a histogram of daily AQI; the toggle switches to log AQI. Raw AQI piles up at 50 to 100 with a long right tail.
- **QQ plot:** compares the data's quantiles with a normal curve. Points on the dotted diagonal = normal. Grey (raw AQI) bends away; blue (log AQI) hugs the line, one reason we model the log.
- **Monthly profile:** average AQI per month, coloured by CPCB category, with ± one standard deviation. January averages 168 (Moderate) and August 52 (Satisfactory). The Kruskal-Wallis $p < 0.001$ confirms months differ.
- **Weekday profile:** average per weekday. Kruskal-Wallis $p = 0.99$: weekdays do not differ, so the weekly cycle is weak.

- **Persistence scatter:** each dot is today's AQI against yesterday's. r_1 = 0.91 (today is very close to yesterday), r_7 = 0.76, r_{365} = 0.65 (same day last year is still related).
- **Prominent pollutant:** which pollutant drove the AQI most often; PM10 and PM2.5 dominate.
- **Outlier inspection:** 21 days flagged in red where AQI jumped far above its 30-day local level (robust z-score above 3.5). They are kept, because they are real pollution events, not errors.

03 Missing values

- **Headline numbers:** 4.6% of days (173) are missing across the full history, the longest gap is 19 days, but only 5 days are missing in the 2019 to 2025 modelling window.
- **Missing days per year:** most gaps are in 2015 to 2018.
- **Imputation comparison:** we hid real values in random 1 to 3 day blocks and rebuilt them three ways. Time interpolation on the log scale was most accurate (average error 14.1 AQI) against forward fill (19.3) and same-weekday-last-week (29.1). It is only used for gaps of 3 days or less.

04 Stationarity

A stationary series keeps the same average and spread over time, which AR/MA-type models assume.

- **Rolling chart:** the 30-day rolling mean (black) and rolling standard deviation (blue). Both rise every winter, so spread grows with level.
- **Test table:** for each version of the series, the ADF test (null hypothesis: non-stationary) and the KPSS test (null: stationary). Log AQI: ADF p = 0.009 rejects non-stationarity and KPSS p ⩾ 0.10 does not reject stationarity, so both agree it is stationary around its seasonal pattern. The verdict chip summarises the pair.

05 Transformations

- **Box-Cox λ = 0.03.** Box-Cox finds the power transform that best stabilises variance; λ near 0 means a log transform.
- **corr(mean, std) 0.81 to 0.29:** month by month, spread tracks the level on the raw scale but much less after the log.
- **Rolling chart of log AQI:** the spread is now far more stable. Every model is fitted on log AQI and converted back with exp(), which gives the median AQI; bands are converted the same way and capped at 500.

06 Seasonality and decomposition

- **Strength scores (0 = none, 1 = dominant):** annual 0.78, weekly 0.05, trend 0.18. The yearly winter-summer cycle drives Mumbai's AQI; the week barely matters.
- **MSTL chart:** five stacked panels that add up to log AQI: *observed*, *trend* (slow drift across years), *annual* (the yearly cycle; the black curve is its smooth 3-harmonic fit), *weekly* (tiny 7-day ripple), *residual* (what is left).
- **Classical monthly decomposition:** each bar is a month's seasonal factor. January is ×1.72 the yearly level and August ×0.48 (red = above average, blue = below).

07 ACF and PACF

- **ACF bars:** correlation between today and k days ago, for k = 1 to 40. **PACF bars:** the same, after removing the effect of the days in between. Blue bars cross the dotted significance lines (±1.96/√n); grey bars do not.
- **Buttons** switch the series: log AQI, its daily change, and the MSTL residual.
- **Reading for log AQI:** the ACF fades slowly while the PACF is significant at lags 1, 3 to 6 and 8 to 10, which points to an autoregressive model of about 10 lags. After differencing the lag-1 ACF is close to zero (−0.07).
- **400-lag ACF:** a wave peaking near lag 365 (red line), the annual cycle seen through autocorrelation.

08 Model comparison

- **Model table:** the idea and assumptions behind each of the 9 models.
- **Protocol note:** how models were chosen on 2024 and scored on 2025 without seeing it.
- **Test set accuracy table:** one row per model, switchable to 1, 3 or 7 days ahead; the selected model is highlighted and the best value per column is bold. Hovering a term in the glossary chips explains it.
- **"Is the gain real?" note:** reads the table for you and reports the Diebold-Mariano result.
- **RMSE by horizon:** horizontal bars per model for 1, 3 and 7 days. Errors grow with the horizon for every model.
- **Validation, 2024:** the scores used to pick the final model.
- **Actual vs predicted, 2025:** the black line is what happened, blue is what the model said 1, 3 or 7 days earlier, the band is its 95% interval.

For AR(10) on Mumbai: MAE 10.3 / 15.8 / 19.0 AQI at 1 / 3 / 7 days, 95% coverage 97% / 96% / 94%, and the CPCB category was right on 84% / 77% / 73% of days.

09 Lag selection

- **Selected orders table:** the winning order in each family with its parameter count, AIC, BIC and why that range was searched. Mumbai: AR(10), MA(10), ARMA(3,3), ARIMA(2,1,3), SARIMA(3,0,3)(0,0,1)[7], SARIMAX with 3 AR lags and 2 annual harmonics.
- **AIC and BIC vs order charts:** for AR and MA, how the scores change as lags are added. The lowest point is the best balance of fit and simplicity. BIC penalises extra parameters harder, so it can prefer fewer lags. The MA curve keeps falling, a sign a pure MA model cannot capture this much persistence.

10 AIC and BIC

- **Full candidate table:** all 75 models fitted on 2019 to 2023, filterable by family and sortable by clicking a column. Highlighted rows were selected. Lower AIC/BIC is better, but only compare rows with the same *d* (differencing changes what the likelihood measures). The final choice also used out-of-sample error, because the lowest AIC does not guarantee the best forecast.

Residual diagnostics

A good model leaves residuals (its one-step errors) that look like random noise.

- **Ljung-Box at lags 7, 14, 28:** tests whether residuals are still correlated. Green chip ($p \geq 0.05$) = no leftover pattern. AR(10) passes all three ($p = 0.36, 0.48, 0.80$).
- **Jarque-Bera:** tests whether residuals are normal. Red for every model: excess kurtosis 1.25 means more extreme days than a bell curve, so intervals can miss the worst spikes.
- **Residuals over time, residual ACF, residual QQ plot:** the same checks as pictures. Red ACF bars mark lags that are still significant.

Limitations

The last section lists what the models cannot do: no weather or festival inputs, linear models on log AQI, a changing station network, and accuracy tested only to 7 days.

Glossary

Term	Meaning
AQI	CPCB National Air Quality Index, 0 to 500; city value = average over its stations
MAE	Mean absolute error: average miss in AQI points

Term	Meaning
RMSE	Root mean squared error: like MAE but big misses count more
MAPE / sMAPE	Average percentage error (sMAPE is the symmetric version)
MASE	MAE divided by the naive model's one-day training MAE; below 1 beats naive. It rises above 1 at longer horizons by design
Coverage 80 / 95	Share of real days inside the 80% / 95% band; ideally near 80% and 95%
Category accuracy	Share of days where the predicted CPCB category was right
DM p	Diebold-Mariano test vs naive; below 0.05 = significantly more accurate
ADF / KPSS	Stationarity tests with opposite null hypotheses, used together
ACF / PACF	Correlation with past days, without / with the in-between days removed
AIC / BIC	Fit scores that penalise extra parameters; lower is better
p, d, q	AR lags, number of differences, MA lags in ARIMA(p,d,q)
[7]	A seasonal term repeating every 7 days
Fourier K	Number of sine/cosine pairs used to draw the yearly cycle

How to run it, and what's next

Install Python 3.11 or newer. Get the code from github.com/Anumay1231/aqi-forecasting-lab (Code, then Download ZIP), then double-click `run.bat` on Windows or run `./run.sh` on macOS/Linux. In the shared project folder, `Run AQI Dashboard.bat` does the same. The first launch installs the packages in a minute or two; the dashboard then opens at http://localhost:8000.

Next steps, if there is time: add weather inputs (wind, rainfall) to explain Mumbai's sudden swings, and add a Diwali and crop-burning event flag for Delhi.

Data: CPCB daily AQI bulletins, compiled by UrbanEmissions.info (GPL-3.0).